

A Percent Needs Its Base

AP Statistics · Unit 1 · Topic 1.3 · B: 2026-09-11 · E: 2026-09-14 · TEACHER KEY

CED TETHER

- **Skill 2.B** — Construct numerical or graphical representations of distributions.
- **Skill 2.A** — Describe data presented numerically or graphically.
- **Enduring Understanding UNC-1** — Graphical representations and statistics allow us to identify and represent key features of data.
- **LO UNC-1.A** — Represent categorical data using frequency or relative frequency tables. [Skill 2.B]
- **EK UNC-1.A.1** — A frequency table gives the number of cases falling into each category. A relative frequency table gives the proportion of cases falling into each category.
- **LO UNC-1.B** — Describe categorical data represented in frequency or relative tables. [Skill 2.A]
- **EK UNC-1.B.1** — Percentages, relative frequencies, and rates all provide the same information as proportions.
- **EK UNC-1.B.2** — Counts and relative frequencies of categorical data reveal information that can be used to justify claims about the data in context.

Essential question: When does a count answer the question, and when does a percent — and what does each one hide?

DOK-3 skill: 4.B · **FRQ pattern:** frequency-vs-relative-frequency-which-table-the-argument-needs

DOK rationale: Part (c) requires matching a stated question to the table that can answer it (counts vs. relative frequencies), citing the numbers that settle it, and reasoning about what each club's chosen table conceals from a reader; the choice must be defended against the other table rather than read off, and either table can be argued for.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK: 1**
→ 3 **Minutes: 45**

Teacher does

- Board slide up at the bell. Read the council's question aloud once; do not say which table you prefer.
- Video follow-along from minute 5 (frequency vs. relative-frequency tables).
- Board slide back up at minute 33. Circulate; every time a student says a percent, ask 'of how many?'
- Collect at the door. Score part (c) only, E/P/I. Either table can earn E if the reason is tied to the question.

Students do

- One-sentence first take with one number, before any teaching.
- Video follow-along (build and read frequency and relative-frequency tables).
- Finish (a)–(c) with the rules callout in front of them; exit line; turn in.

Questions to ask

- If the session room holds 30 chairs, which table tells the council how many chairs each club fills?

- 83% of what? 60% of what? Say the base out loud.
- Why did Robotics choose counts and Chess choose percents? What would each look like on the other table?
- Rewrite the council's question so that the *other* table is the right one to answer it.

Adult role

Solo teacher. Circulate 5 pairs. Do not name a table yourself during the finish window; ask what the question is asking for — people or share.

[WARN] WATCH FOR

- 'Percents are always fairer' stated as a rule, with no link to what the council actually asked.
- Citing 83% without ever saying it is 15 of 18 people.
- Comparing 27 Yes to 15 Yes as if the clubs were the same size.

SCORING PART (c) — E / P / I

Expected elements

- **table-tied-to-question** (required) — Commits to one table and ties the choice to what the council's question asks (how many students vs. what share of a club); either table accepted with that reason
- **cites-settling-numbers** (required) — Cites the specific numbers that settle it (27 vs. 15 Yes, or 83% vs. 60%)
- **counts-hide-base-sizes** (required) — Says the Robotics frequency table hides that the clubs are different sizes (45 vs. 18), so 27 Yes is only 60% / 18 said No
- **percents-hide-small-n** (required) — Says the Chess percent table hides that 83% is 15 people out of only 18 (a percent needs its base)

E	Commits to one table with a reason tied to the council's question, cites the settling numbers, AND names both hides (club sizes behind the counts; the small base behind the percent).
P	Commits with numbers but names only one hide, OR names both hides but never commits to a table or gives no reason tied to the question, OR the reason is 'percents are always fairer' with no link to what was asked.
I	Says the two tables say the same thing, or picks a table with no numbers, or names no hide.

Common mistakes

- 'Percents are always better' (or 'counts are the real data') with no link to what the council asked
- Cites 83% without ever saying it is 15 people out of 18
- Compares 27 Yes to 15 Yes without noting Robotics has 45 members to Chess's 18
- Reports 27/45 as '27%' — treats a count as a percent
- Rounds 15/18 to 80% or 85%

[STAR] TODAY'S PROBLEM — annotated

Two clubs asked the student council for money to run a Saturday session. Each club surveyed *all* of its members: “Would you attend a Saturday session?” The table shows every response as a count.

At the council meeting, the **Robotics club** presented a **frequency table** (counts) and the **Chess club** presented a **relative-frequency table** (percents). Each club chose the table that made its case look best.

“Would you attend a Saturday session?” (every member surveyed; counts)

Club	Yes	No	Total
Chess	15	3	18
Robotics	27	18	45

FIRST TAKE (5 min)

Which club has more interest in a Saturday session — Chess or Robotics? One sentence and one number.

Key: Not graded. Expect a split: some say Robotics (27) and some say Chess (83%); the split IS the lesson.

Rules callout “TABLES FOR A CATEGORICAL VARIABLE (keep on the page)” is printed on the student sheet.

DOK 1 (a) How many Robotics members said Yes? Which club has more members, and how many more?

Key: 27 Robotics members said Yes. Robotics has more members: 45 vs. 18, so 27 more.

DOK 2 (b) Build the relative-frequency table: for each club, the percent of its members who said Yes and the percent who said No (nearest whole percent). Which club has the higher percentage saying Yes?

Key: Chess: Yes $15/18 \approx 83\%$, No $3/18 \approx 17\%$. Robotics: Yes $27/45 = 60\%$, No $18/45 = 40\%$. Each row adds to 100%. Chess has the higher percentage saying Yes (83% vs. 60%).

DOK 3 (c) The council asks: “Which club has **more student interest** in a Saturday session?” Decide which table — counts or percents — answers *that* question, and cite the specific numbers that settle it. Then explain what each club’s chosen table **hides** from a reader: what does the Robotics club’s frequency table hide, and what does the Chess club’s percent table hide?

Key: Expected: “more student interest” asks *how many students* are interested, so the **frequency table** answers it: 27 Robotics members said Yes vs. 15 Chess members — 12 more people would show up. (A student who reads the question as “which club’s *members* are more interested” and picks the percent table, 83% vs. 60%, earns full credit *if they say that is the question the percents answer*.) What the tables hide: the Robotics club’s **counts** hide the base sizes — Robotics has 45 members to Chess’s 18, so 27 Yes is only 60% of the club and 18 Robotics members said No. The Chess club’s **percents** hide how small the base is — 83% is 15 people out of only 18; 83% of 18 (15) is fewer students than 60% of 45 (27).

EXIT

One thing the video changed about my first take: _____